

Problem Statement No.18 - AI Agent for Electricity Bill Fraud Detection

The Challenge

Electricity fraud and billing anomalies lead to major financial losses for power distribution companies. With thousands of consumers and massive billing data, detecting suspicious consumption patterns manually is extremely difficult. Utilities need an intelligent AI system that can automatically analyze electricity usage data and identify potential fraud cases.

The Objective

Build an **AI Agent for Electricity Bill Fraud Detection** that analyzes electricity consumption and billing records to detect abnormal patterns and flag potential fraud cases.

The solution should:

1. **Electricity Consumption Analysis** – Analyze consumer electricity usage data to understand normal consumption behavior and trends.
2. **Fraud Pattern Identification** – Detect suspicious activities such as sudden drops in usage, unusual spikes, or inconsistencies between meter readings and billed units.
3. **AI-Based Anomaly Detection** – Automatically identify consumers whose usage patterns significantly deviate from normal historical behavior.
4. **Explainable AI Insights** – Provide clear explanations for why a particular consumer or electricity bill is flagged as suspicious.

Tech Stack (Mandatory)

- IBM Langflow / IBM Orchestrate + IBM Models